

Tutorial 5**Determinants, eigenvalues and eigenvectors**

1. Let A be a 7×7 matrix, with $\det(A) = 3$. Find
(a) $\det(-4A)$ (b) $\det(A^{-1}A^T)$ (c) $\det(A^4)$ (d) $\det(AA^T)$.

2. Let

$$A = \begin{pmatrix} 3 & 0 & -9 & 6 \\ -2 & 5 & 1 & -1 \\ -1 & 7 & 4 & -2 \\ 4 & -1 & 2 & 3 \end{pmatrix}.$$

- (a) Perform elementary row operations on matrix A to make all the entries in the first column, except the first entry, to zero.
(b) Use (a) to find $\det(A)$.
3. Prove or disprove:
(a) For any square matrix A , $\det(-A) = -\det(A)$.
(b) If A is a square matrix, then $\det(A^T A) \geq 0$.
(c) If A and B are square matrices of the same size, $\det(A + B) = \det(A) + \det(B)$.
(d) If A is a square matrix such that $A^T = A^{-1}$ then $\det(A) = 1$ or $\det(A) = -1$.

4. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 2 \\ 1 & 2 & 3 \end{pmatrix}$.

- (a) Find all the eigenvalues of A .
(b) For each eigenvalue you found in (a), find a corresponding eigenvector.
[Hint: For each eigenvalue, you have to solve a system of linear equations to find a corresponding eigenvector.]

5. Let $A = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{pmatrix}$.

- (a) Write down the determinant you have to evaluate to find the eigenvalues of A (there will be λ 's in the determinant).
 - (b) Perform a row operation on this determinant to obtain a zero in the first column.
 - (c) Then make use of the properties of determinants to find the value of the determinant, in terms of λ .
 - (d) Now find all the eigenvalues of A .
 - (e) For each eigenvalue you have found, find a corresponding eigenvector.
6. Let A and P be $n \times n$ matrices, with P invertible.
- (a) Show that A and $P^{-1}AP$ have the same eigenvalues.
[*Hint:* $I = P^{-1}IP$.]
 - (b) If λ is an eigenvalue of A and $\mathbf{x}$ a corresponding eigenvector of A , is it also an eigenvector of $P^{-1}AP$? If not, find an eigenvector of $P^{-1}AP$ corresponding to λ .